Final Year Project: Fish Disease Detection System

1. Introduction

For your Final Year Project, you are presenting 'AquaVision AI', a comprehensive system designed to detect diseases in freshwater fish using Deep Learning. The project addresses the critical issue of disease outbreaks in aquaculture which cause significant economic losses.

2. Problem Statement

- Manual disease detection is slow, error-prone, and requires experts.
- Late detection leads to rapid disease spread and high mortality.
- Lack of accessible tools for farmers to diagnose fish health on-site.

3. Project Objectives

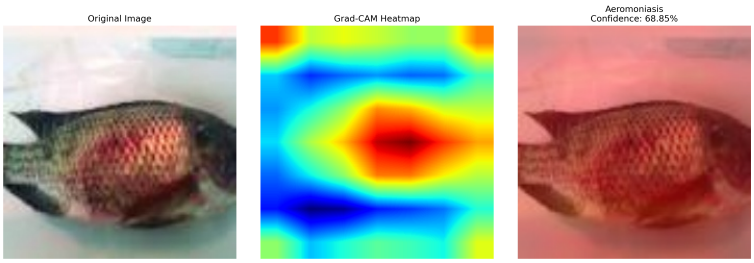
- Develop an accurate AI model (>75% accuracy) to classify 7 distinct fish conditions.
- Implement 'Explainable AI' (Grad-CAM) to visualize disease symptoms, building trust.
- Create a user-friendly Desktop Application for real-time diagnosis.
- Ensure the system can run offline on edge devices (low latency).

4. Methodology & Technology

We utilized Transfer Learning with the MobileNetV2 architecture, pre-trained on ImageNet, fine-tuned on our dataset of 2,400+ fish images.

Key Technologies:
- Framework: TensorFlow/Keras
- GUI: CustomTkinter (Python)
- Database: SQLite (for tracking history)
- Interpretability: Grad-CAM (Gradient-weighted Class Activation Mapping)

Figure 1: Grad-CAM Visualization (AI "Vision")



The image above demonstrates the system's explainability. The heatmap highlights the specific regions of the fish (e.g., gills, tail) that the AI used to make its diagnosis, helping users verify the result.

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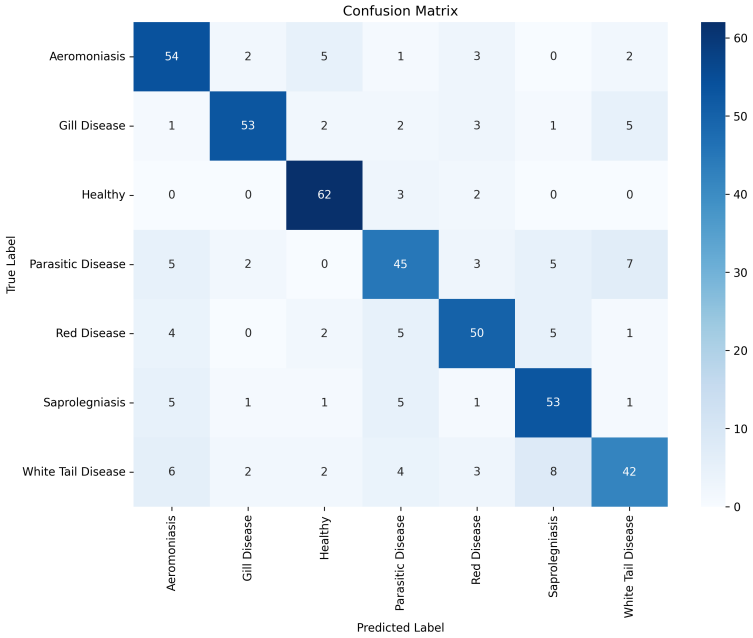
5. Performance Results

The model achieved strong performance metrics on the test set:
- Overall Accuracy: 76.55%
- Precision: 88.42% (Minimizing false alarms)
- Inference Speed: <1 second per image (Real-time capability)
- Model Size: 2.40 MB (Optimized for deployment)

Figure 2: Confusion Matrix



6. Conclusion

AquaVision AI successfully automates fish disease detection. It empowers farmers with expert-level diagnostics without the need for constant internet connectivity. The inclusion of Grad-CAM transparency and a robust history tracking system makes it a professional-grade tool suitable for real-world aquaculture applications.

7. Presentation Tips

- Live Demo: Show the 'Real-time Camera' feature during your presentation if possible.
- Highlight user benefits: 'Saves money', 'Saves time', 'Easy to use'.
- Emphasize 'Explainable AI': Judges love knowing *why* the AI made a decision.